Cell test bench — schematic draft v0.1 (generated). One 18650/21700 cell at a time; 0–10 A load, <=5 A charge, Peltier-controlled cell pocket.

Inter-sheet connections use global labels. Power nets: GND, +12V, +5V, +3V3 (Blue Pill), VIN (30 V), +12V_SW (after thermostat).

Power

Power input and rails

File: power.kicad_sch

Cell_Relays

Cell connection, shunt and relays K1/K2

File: relays.kicad_sch

Electronic_Load

Electronic load 0–10 A (linear, CC)

File: load.kicad_sch

Charger

Charger: XL4016 CC/CV module with DAC-set voltage

File: charger.kicad_sch

Measurement

Measurement: 2 x ADS1115, NTCs, fast voltage

File: measurement.kicad_sch

Protection

Hardware protection (independent of firmware)

File: protection.kicad_sch

Thermal

Peltier driver and fans

File: thermal.kicad_sch

MCU

Blue Pill (STM32F103C8) and interfaces

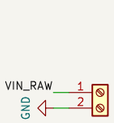
File: mcu.kicad_sch

Solaris — cell test bench

Sheet: /
File: cell_test_bench.kicad_sch

Title: Cell test bench

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/9



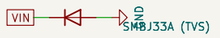
J101

PSU in 30V 10A (+, –)



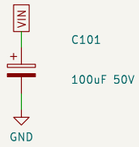
F101

10A



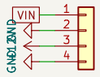
D101

SMBJ33A (TVS)



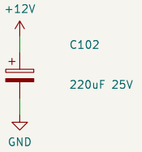
C101

100uF 50V



J102

12V buck module: IN+ IN– OUT+ OUT–



C102

220uF 25V



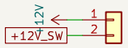
J103

5V buck module: IN+ IN– OUT+ OUT–



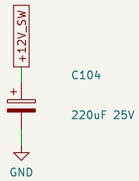
C103

100uF 16V



J104

Thermostat KSD9700 60C NC



C104

220uF 25V











PSU: existing 30 V / 10 A bench supply, no CC/CV. F101 + D101 (TVS) protect against overvoltage and reverse polarity (D101 conducts, F101 blows).

J102: 12 V buck module (XL4015 or LM2596HV) for Peltiers, relays and fans. J103: 5 V buck module (MP1584) for Blue Pill, comparators and BTS7960 logic.

+3V3 comes from the Blue Pill regulator (MCU sheet).

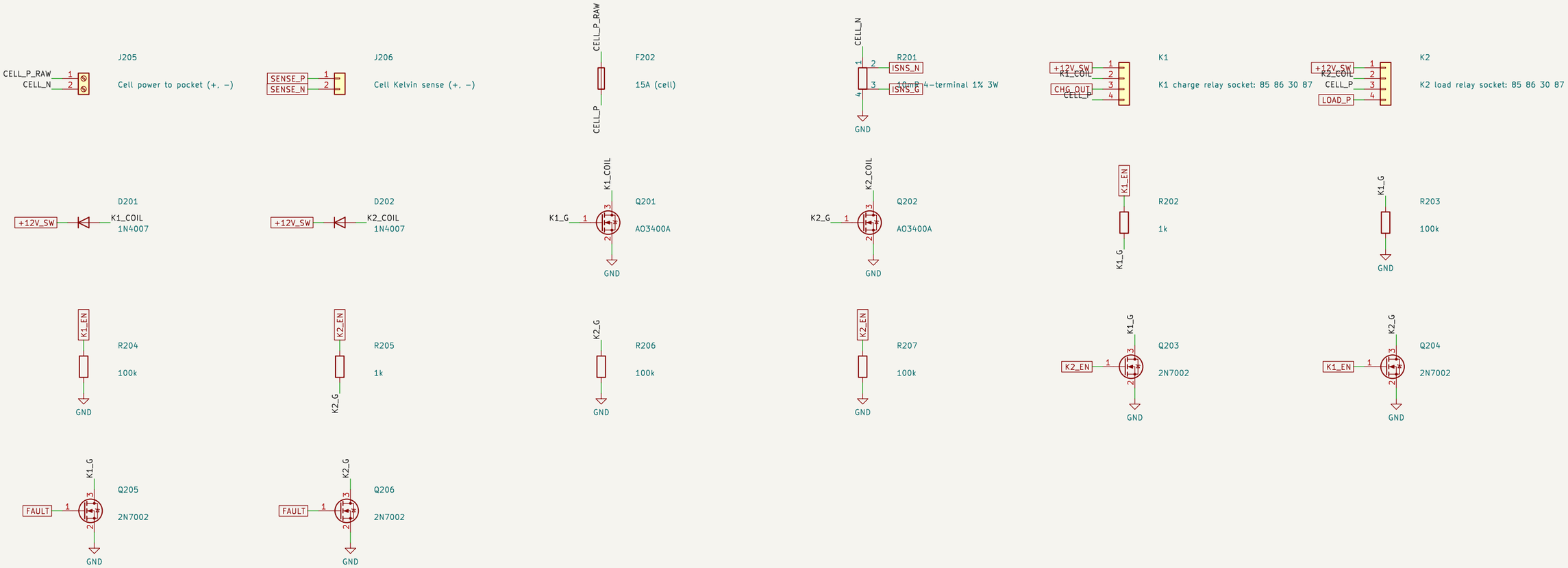
J104: bimetal thermostat KSD9700, normally closed, 60 C, mounted on the load heatsink. It cuts +12V_SW (relay coils + Peltier driver) independently of firmware.

Solaris – cell test bench

Sheet: /Power/
File: power.kicad_sch

Title: Power input and rails

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/9



Single 10 mOhm 4-terminal shunt R201 sits in the cell negative lead, so it measures BOTH charge and discharge current. System GND = shunt ground side (star point).

Kelvin sense: ISNS_N / ISNS_G from the shunt sense pins, SENSE_P / SENSE_N from separate wires at the cell tabs.

K1 (charger) and K2 (load): 12 V automotive relays 30–40 A, normally open, in sockets (pins 85, 86, 30, 87).

Interlock: Q203 holds K1 off while K2_EN is high, Q204 holds K2 off while K1_EN is high (both commanded = both off). Q205/Q206 drop both relays on FAULT.

R204/R207 pull-downs keep both relays off while the MCU boots.

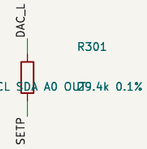
Solaris — cell test bench

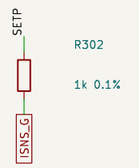
Sheet: /Cell_Relays/
File: relays.kicad_sch

Title: Cell connection, shunt and relays K1/K2

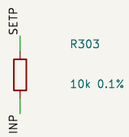
Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/9

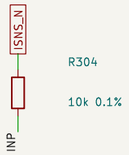


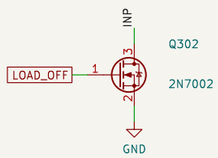


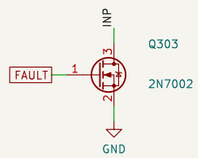


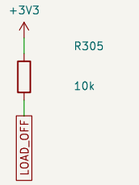


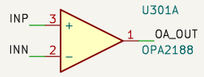


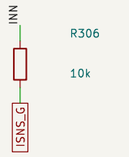




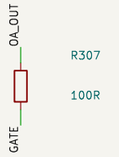


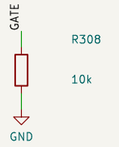


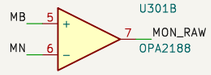


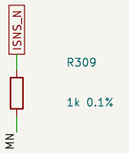


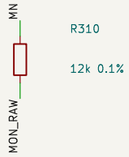


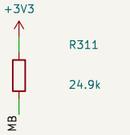


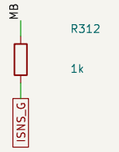


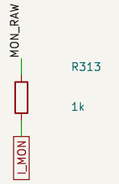


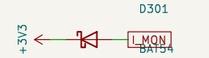




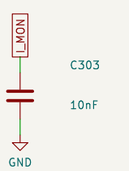


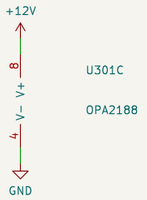


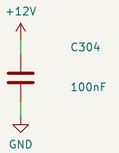


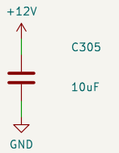












CC loop: U301A compares the DAC setpoint with the shunt voltage. Summing node INP = (SETP − I*Rshunt)/2, so the loop settles at I = SETP / 10 mOhm.

Setpoint: MCP4725 #1 (0x60) 0–3.3 V, divided by R301/R302 to 0–108 mV -> 0–10.8 A full scale (~2.6 mA per LSB).

Q301 IRFP260N in linear mode on the CPU-cooler heatsink, with the external 0.1 Ohm 50 W aluminium resistor (J301) in series to share the heat.

Kill: Q302 (LOAD_OFF from MCU, pulled up = off at boot) and Q303 (hardware FAULT) short INP to ground -> gate goes low.

Sequence: close K2 with LOAD_OFF high, release LOAD_OFF with DAC = 0, then ramp the DAC.

U301B: bidirectional current monitor for the fast internal ADC and the OC comparators: I_MON = 1.65 V + 0.12 V/A (approx −13 A … +13 A).

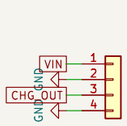
U301 = OPA2188 (zero-drift, 36 V, input range includes V−). Check pinout of the DAC module you buy (order varies).

Solaris — cell test bench

Sheet: /Electronic_Load/
File: load.kicad_sch

Title: Electronic load 0–10 A (linear, CC)

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/9



J401

XL4016 CC/CV module: IN+ IN− OUT+ OUT−



J402

XL4016 FB tap (wire to FB pin, GND)



U401

MCP4725 module #2 (A0=VCC, 0x61): VCC GND SCL SDA A0 OUT

R402

100R



C401

1uF



R401

1.2k



D401

BAT54

Set the module CV pot to 4.25 V with no load BEFORE wiring the FB tap. That pot value is the hardware voltage ceiling.

DAC #2 (0x61) injects current into the XL4016 FB pin (1.25 V) through D401 + R401. Injection can only LOWER the output, so DAC = 0 / MCU dead = 4.25 V ceiling, never higher.

With the module bottom FB resistor ~360 Ohm and top ~860 Ohm: DAC 0 V −> 4.25 V, DAC 3.3 V −> ~3.0 V. Measure your module and adjust R401.

The module own CC shunt is bypassed because IN− and OUT− share system GND. Turn its CC pot to max; current is regulated by firmware (ADS1115 + shunt) and limited by the OC comparator.

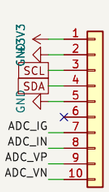
Firmware CC/CV: lower/raise the DAC until measured current = target, then hold 4.2 V and stop at C/20 by opening K1.

Solaris — cell test bench

Sheet: /Charger/
File: charger.kicad_sch

Title: Charger: XL4016 CC/CV module with DAC−set voltage

Size: A3	Date: 2026−10−13	**Rev: 0.1 draft**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 5/9



U501

ADS1115 module #1 (ADDR=GND): VDD GND SCL SDA ADDR ALRT A0 A1 A2 A3

R501

100R



R502

100R

C501

100nF



R503

10k 0.1%



R504

10k 0.1%

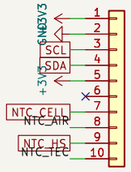


R505

100R

C502

100nF



U502

ADS1115 module #2 (ADDR=VDD): VDD GND SCL SDA ADDR ALRT A0 A1 A2 A3

R506

10k 0.1%



C503

100nF



J501

NTC 10k cell



R507

10k 0.1%

C504

100nF



J502

NTC 10k air

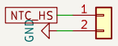
R508

10k 0.1%



C505

100nF



J503

NTC 10k hs



R509

10k 0.1%

C506

100nF



J504

NTC 10k tec

R510

10k



R511

10k



C507

10nF

ADS1115 #1 (0x48): A0–A1 differential = shunt (current, +–256 mV range), A2–A3 differential = cell voltage / 2 (Kelvin, +–2.048 V range).

ADS1115 #2 (0x49): four NTC channels (10k B3950, 10k pull–up): cell surface, pocket air, load heatsink, Peltier hot side.

V_FAST: cell voltage / 2 for the Blue Pill internal ADC (pulse capture at ~1 kHz). It is ground–referenced, so subtract I x shunt in software.

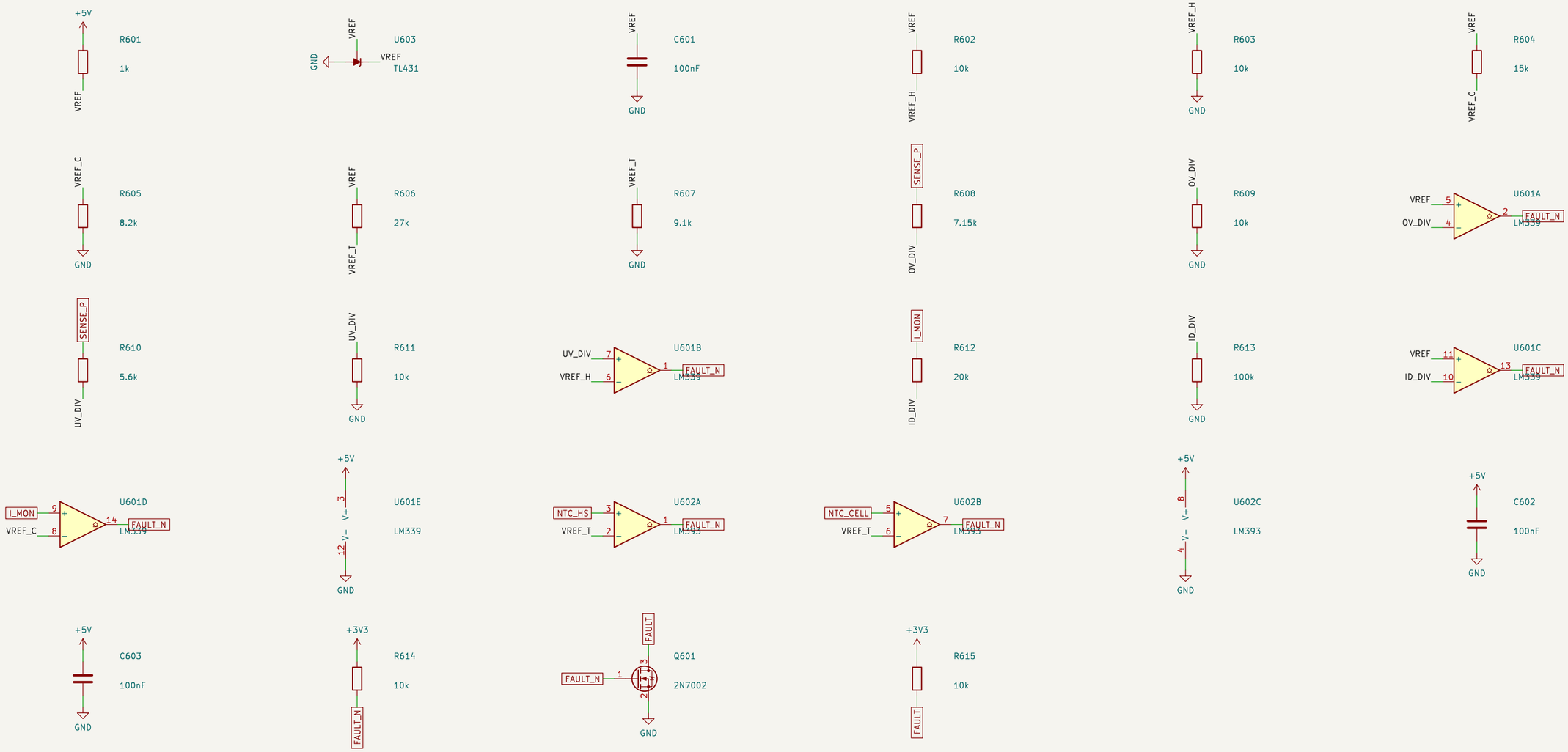
Divider resistors R503/R504 should be 0.1 % 25 ppm, or calibrate against a DMM.

Solaris – cell test bench

Sheet: /Measurement/
File: measurement.kicad_sch

Title: Measurement: 2 x ADS1115, NTCs, fast voltage

Size: A3	Date: 2026–10–13	**Rev: 0.1 draft**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 6/9



VREF = 2.5 V from TL431 (U603). Comparators run from +5 V, open-collector outputs wired-OR onto FAULT_N (pull-up to 3V3).

Trips: OV cell > ~4.29 V | UV cell < ~1.95 V | OC discharge > ~11 A | OC charge > ~6.4 A | heatsink or cell NTC > ~62 C.

Q601 inverts FAULT_N -> FAULT (active high), which kills the load setpoint, both relay gates and the Peltier driver enable.

The MCU reads FAULT_N on PB15 and can also pull it low (open-drain) to force a safe stop and latch the fault in firmware.

UV also trips with no cell inserted (shows as "no cell"). Add ~10 mV hysteresis resistors later if the comparators chatter.

Solaris — cell test bench

Sheet: /Protection/
File: protection.kicad_sch

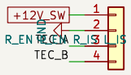
Title: Hardware protection (independent of firmware)

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/9



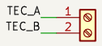
J701

BTS7960 IBT-2 logic: RPWM LPWM R_EN L_EN VCC GND



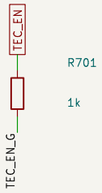
J702

BTS7960 power: B+ B- M+ M-



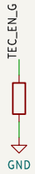
J703

Peltier x2 in series



R701

1k



R702

100k



Q701

2N7002



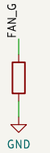
Q702

AO3400A



R703

100R



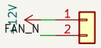
R704

100k



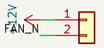
D701

SS14



J704

Fans 12V: heatsink (+,-)



J706

Fans 12V: Peltier hot side (+,-)

J705

Pocket mixing fan 5V

BTS7960 (IBT-2) H-bridge module drives two Peltiers in series from +12V_SW (about 6 V each, better efficiency). RPWM = heat/cool one way, LPWM = the other.

TEC_EN goes to R_EN and L_EN; Q701 pulls it low on FAULT. The thermostat on the Power sheet also removes +12V_SW.

Check that the IBT-2 logic accepts 3.3 V inputs when powered from 5 V; if not, add a 74HCT245 level shifter.

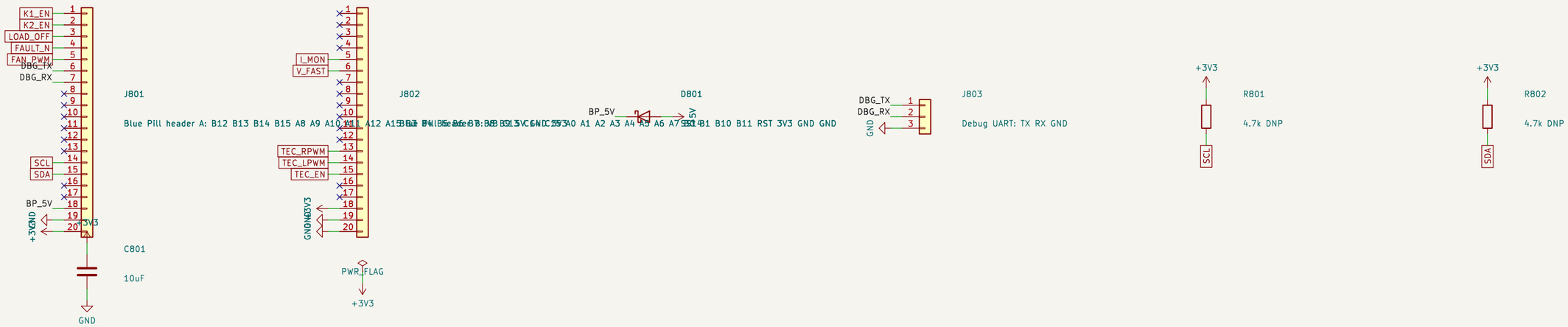
Q702 switches both 12 V fans (load heatsink, Peltier hot side) with PWM from PA8. J705 is an always-on 5 V mixing fan inside the pocket.

Solaris — cell test bench

Sheet: /Thermal/
File: thermal.kicad_sch

Title: Peltier driver and fans

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/9



Blue Pill plugs into two 1x20 female headers. USB (micro-USB on the Blue Pill) goes to the PC: USB CDC for the Python UI.

D801 stops the 5 V rail from back-feeding the PC USB port (the Blue Pill 5V pin is tied to VBUS).

Pins: PA0 I_MON, PA1 V_FAST (ADC) | PB6/PB7 I2C1 | PB12 K1_EN, PB13 K2_EN | PB14 LOAD_OFF | PB15 FAULT_N (open-drain, read + force) |

PB0/PB1 TIM3 TEC_RPWM/LPWM, PB10 TEC_EN | PA8 TIM1 FAN_PWM | PA9/PA10 debug UART | PA11/PA12 USB (on board).

R801/R802: I2C pull-ups. Fit only if the modules do not already have them.

Solaris — cell test bench

Sheet: /MCU/
File: mcu.kicad_sch

Title: Blue Pill (STM32F103C8) and interfaces

Size: A3	Date: 2026-10-13	**Rev: 0.1 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/9